

JIZE LI

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EDUCATION

University of Pennsylvania <i>Master of Computer and Information Science</i>	August 2025 - Now GPA: 4.0
College of William and Mary <i>Bachelor of Science, Major in Computer Science</i> <i>Northrop-Grumman Award (Department Award)</i>	August 2021 – May 2025 Major GPA: 4.0
<i>Bachelor of Science, Major in Economics</i> <i>Terry Glenn Memorial Scholarship (Department Award)</i>	Major GPA: 3.93
Tsinghua University High School <i>Beijing, China</i>	August 2018 – June 2021 GPA: 3.98

SKILLS

Programming Languages: Python, Java, C, JavaScript, SQL
Technologies/Frameworks: HTML, CSS
Spoken Languages: English, Chinese

WORK EXPERIENCE

AI Research & Development Intern <i>Microsoft Cloud and AI Group - Microsoft Research Asia</i>	May 2024 – August 2024 Beijing, China
- Researched on the theory basis of multiple Multimodal Large Language Models including Qwen-VL, Kosmos, Phi3-V, and other Visual Transformer Models like Nougat. - Constructed dataset and conducted supervised finetuning on Qwen-VL-Chat model using QLora to perform figure detection tasks on academic paper. - Researched and compared the performance of Image to Markdown Conversion task between Nougat, Kosmos-2.5, ChatGPT-4o, and Phi3-V. - Constructed dataset and conducted supervised finetuning on Phi3-V model to improve its performance on Image to Markdown Conversion, especially on math formula recognition.	
Software Development Intern <i>Zhongkeqidi Optoelectronic Technology Co., Ltd.</i>	July 2023 – September 2023 Beijing, China
- Wrote a satellite anti-spoof detection algorithm integrated with high precision atomic clock. - Developed a software that tracks and displays the real time clock offset employing the algorithm and window slots to set serial parameters and other options. - The software was later integrated into NTP server to provide anti-spoof service for companies/individuals. The product has been put on sale for commercial use.	

RESEARCH EXPERIENCE

Regression Analysis of the 1973 Oil Crisis' Impact on Nuclear Energy <i>Econometrics</i>	May 2024 Independent Research
- Investigated the impact of the 1973 oil crisis on nuclear-powered electricity generation using panel data of 25 nuclear-capable countries (1960–2010). - Implemented and compared results from two-way fixed effects model, random effects model and pooled OLS. Used panel data and Chow tests to identify shifts in energy strategies. - The results showed the oil crisis as a significant catalyst for increased dependence on nuclear power.	
Time Series Forecasting on Store Sales <i>Machine Learning, Kaggle Online Competition</i>	December 2023 Group Research
- Developed a hybrid time series forecasting model integrating linear regression and XGBoost to predict retail sales across diverse product families and locations. - The model decomposed time series data into trend, seasonality, cycle, and residual components, leveraging machine learning techniques to improve accuracy. - Results demonstrated significant performance in capturing intricate sales patterns, supporting more strategic inventory management and reducing waste.	

COMPLETED PROJECTS

-  **Meeting Website Based on Robert's Rules of Order (RONR)** (Comprehensive Web Application) *TailwindCSS/React/MongoDB* November 2024
- The platform includes a front-end, back-end, and database, providing a structured, user-friendly environment for users to conduct organized meetings with clear procedural guidelines.
 - Users can create accounts, manage profiles, and join or create meetings. Different roles include Chair and Member, each with different permissions and operations within meetings.
 - In a meeting, members make and second motions, debate on motions, vote, and keep track of meeting minutes in real time.
-  **IMG2Markdown Finetune on Phi3-V** (Multimodal Large Language Model) | *Python* August 2024
- The base version of Phi3-V has no ability to read image and generate corresponding Markdown code.
 - Built specialized training dataset with around 10k images with ground truth using Nougat. Conducted supervised finetuning on Phi3-V with the constructed dataset.
 - The finetuned version of Phi3-V performed excellent on generating Markdown codes. Its accuracy in recognizing math formulas exceeds ChatGPT-4o.
-  **Figure Detection on Qwen-VL-Chat** (Multimodal Large Language Model) | *Python* June 2024
- Constructed training dataset using images and ground truth pairs from publaynet dataset.
 - Finetuned Qwen-VL-Chat model with 16k samples
 - The finetuned version of Qwen model could recognize figures and frame them with bounding boxes. The accuracy rate increased to 40% from less than 5% before training.
-  **Maze Game** (Windows/MacOS App) | *Java* December 2022
- Use **DFS, Prim's Algorithm or Boruvka's Algorithm** to randomly generate mazes with different size, adjusted according to user input.
 - Game starts with maps and solutions, turned off by default.
 - The maze could be deployed with **robots and drivers** that automatically drove the robot to the exit using various pathfinding algorithms. Robots have sensors to gain information and could run out of power and fail.
 - The game was then adapted to android applications, accommodated with new User Interface and background threads.

EXTRACURRICULAR

- Video Content Creator on Bilibili* *Profile Link: <https://b23.tv/vefkztn>*
- Created high-intensity **action AMVs**, blending anime fight scenes into seamless montages.